

MentorMAMA

A digital mentorship toolkit for safer midwifery clinical placement

Developer Concept Note & MVP Specification

Prepared for AfyaVentures / JHUB Africa

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Source basis

This concept is derived from the proposal titled “Developing and Evaluating a Digital Capacity-Building Model for Midwifery Clinical Mentors to Improve Student Mentorship in Maternity Units in Selected County Referral Hospitals in Kenya” by Dr. Carolyne Kerubo Nyariki, School of Nursing, College of Health Sciences, JKUAT.

Item	Developer-facing detail
Product category	Maternal & Reproductive Health; Digital Health; Health Workforce Strengthening
Primary problem	Clinical mentorship for nursing and midwifery students in maternity units is often informal, inconsistent, hard to monitor, and constrained by high student-to-mentor ratios.
Product solution	A mobile-first platform that standardizes mentor training, student onboarding, labour ward induction, mentorship logging, feedback, and monitoring dashboards.
First users	Clinical mentors, nurse managers, midwifery students, university placement coordinators, and AfyaVentures/JHUB program admins.
MVP approach	Build a simple web/PWA platform first. Avoid direct patient care functions and avoid collecting patient-identifiable data in the MVP.
Pilot scope	One or two maternity units linked to JKUAT clinical placement sites; 10-25 mentors; 30-80 students; 8-12 weeks minimum pilot.

1. Product Concept Summary

MentorMAMA is a digital mentorship support tool that helps maternity units provide structured, consistent, and measurable clinical mentorship to nursing and midwifery students during clinical placement.

The platform converts the research proposal into a developer-ready product: a workflow system for onboarding students, training mentors, tracking mentorship activity, collecting student feedback, and producing dashboards for universities, facilities, and program managers.

One-sentence product pitch

MentorMAMA helps maternity clinical mentors guide students better by combining short digital mentor training, a labour ward induction checklist, mentorship session logs, student feedback, and real-time dashboards.

2. Why This Product Is Needed

Clinical placement is one of the most important parts of midwifery education because students learn clinical judgment, respectful maternity care, documentation, teamwork, communication, and management of obstetric conditions in real service settings.

However, the source proposal identifies major gaps in County Referral Hospitals across Nairobi, Kiambu, Murang'a, and Machakos: mentorship is often informal and unstructured, many mentors lack formal mentorship or preceptorship training, and high student numbers limit individualized supervision.

MentorMAMA addresses these gaps by making mentorship visible, structured, and easier to support without adding heavy administrative burden to busy maternity teams.

3. Product Vision and Design Principles

Principle	Meaning for developers
Mobile-first	Most users should be able to use the system from a smartphone. Do not assume desktop access.
Low-bandwidth by design	Pages must load quickly, forms should save drafts, and essential tasks should work even with weak connectivity.
Mentor-centered	The mentor experience must be fast and practical. Session logs and checklists should take minutes, not hours.
Student-safe	Students need a confidential feedback channel and a clear escalation pathway for serious concerns.
Facility-integrated	The system should fit normal maternity unit workflows rather than creating a parallel reporting burden.
No patient-identifiable data in MVP	The first version should track mentorship and learning, not patient records.
Evidence-ready	The platform should produce clean exportable data for monitoring, evaluation, and AfyaVentures impact reporting.

4. Target Users and Roles

User role	Core needs	Main permissions
Student	Clear orientation, assigned mentor, learning expectations, session history, feedback submission.	View own placement, complete onboarding, submit feedback, view mentor assignment.

Clinical mentor	Quick guidance on mentorship, checklist tools, session logging, feedback prompts.	Complete training, access assigned students, complete induction checklist, log mentorship sessions.
Nurse manager / ward in-charge	Visibility on student onboarding, mentor participation, issues, and supervision gaps.	View facility dashboard, assign mentors, review escalated issues, export facility reports.
University coordinator	Monitor student placement quality across facilities and cohorts.	Create cohorts, upload/approve students, view dashboards, export university reports.
Program admin	Configure facilities, users, modules, indicators, and reports.	Full administrative configuration and analytics access.
CPD/content reviewer	Review training content and confirm mentor module completion.	Upload/review content, view mentor completion reports if approved.

5. MVP Scope

The first build should focus on the minimum set of features needed to test whether structured digital mentorship improves mentor engagement, student onboarding, and visibility of clinical learning support.

5.1 In Scope for MVP

- User registration and role-based login for students, mentors, nurse managers, university coordinators, and admins.
- Facility, ward/unit, cohort, placement period, and mentor assignment setup.
- Digital mentor training modules with completion tracking and short knowledge checks.
- Labour ward induction checklist for student onboarding.
- Mentorship session log for mentors.
- Student feedback and confidence tracker.
- Escalation flag for serious placement concerns.
- Dashboards for key mentorship indicators.
- CSV/Excel export for monitoring and reporting.
- Basic notifications by email and optional WhatsApp/SMS integration later.

5.2 Out of Scope for MVP

- Electronic medical records or patient management features.
- Clinical decision support for patient care.
- Collection of patient names, phone numbers, IDs, or medical record numbers.
- Payment, billing, and subscription management.
- Complex AI functionality before the workflow is validated.
- Full native Android/iOS applications unless the team decides this is essential after testing the PWA.

6. Core Product Modules

Module	Description
1. Account and Access Management	Role-based accounts for students, mentors, nurse managers, university coordinators, and admins. Include password reset, account status, and permission boundaries.
2. Facility and Placement Setup	Admins/coordinators create facilities, wards/units, placement cohorts, dates, student groups, and mentor-student assignments.
3. Mentor Training Hub	Short case-based modules on student mentorship, preceptorship, feedback, respectful maternity care, documentation, escalation, and learner support.

4. Labour Ward Induction Checklist	A structured checklist completed when a student reports to the unit. It confirms orientation to workflow, safety, reporting lines, emergency protocols, respectful maternity care, documentation, and learning objectives.
5. Mentorship Session Log	A lightweight form mentors complete after mentorship activities. It records date, student, topic, type of supervision, skills discussed, feedback given, and follow-up actions.
6. Student Feedback and Confidence Tracker	Students submit short feedback on induction quality, mentor availability, confidence, safety of the learning environment, and unresolved support needs.
7. Escalation and Support	Students and mentors can flag serious concerns. Nurse managers/coordinators can review, assign status, document action, and close the issue.
8. Dashboard and Reports	Visual indicators showing onboarding status, mentor training completion, session frequency, student feedback, issues, and cohort progress.
9. Content Library	PDFs, quick guides, short videos, checklists, and facility-specific guidance accessible by role.
10. Admin Configuration	Admin can create/edit modules, indicators, checklist items, facilities, wards, cohorts, and exports.

7. Key User Workflows

7.1 Student Journey

1. Student receives placement invitation or link from university coordinator.
2. Student creates profile and confirms facility/ward placement.
3. Student views assigned mentor and placement expectations.
4. Student completes digital orientation information before arrival.
5. On arrival, mentor or nurse manager completes labour ward induction checklist.
6. During placement, student receives mentorship sessions and can submit feedback.
7. At end of placement, student completes final experience survey and confidence self-assessment.

7.2 Clinical Mentor Journey

8. Mentor receives account invitation and completes mentor profile.
9. Mentor completes digital training modules and knowledge checks.
10. Mentor views assigned students and placement dates.
11. Mentor completes induction checklist for each student.
12. Mentor logs mentorship sessions using a short structured form.
13. Mentor reviews pending follow-ups and unresolved student support needs.
14. Mentor receives completion summary for CPD or facility reporting, subject to approval.

7.3 Nurse Manager / Ward In-Charge Journey

15. Manager verifies active students and mentors in the unit.
16. Manager assigns or reassigns mentors when staffing changes.
17. Manager monitors induction completion and mentorship activity.
18. Manager reviews escalated concerns and records action taken.
19. Manager exports monthly mentorship summary for facility and university coordination.

8. Data Capture Requirements

Data protection rule for MVP

Capture only mentorship and learning data. Do not collect patient-identifiable data. Any open-text fields must clearly instruct users not to enter patient names, phone numbers, medical record numbers, or other identifiers.

Data object	Important fields	Notes
Institution	Name, type, contact person, county, active/inactive.	University or training institution.
Facility	Facility name, county, level, maternity unit name, contact person.	Pilot facilities can be added by admin.
Ward / Unit	Unit name, shift pattern, maximum students, nurse manager.	Useful for staffing and placement tracking.
User	Name, role, phone/email, institution/facility, status.	Use role-based access control.
Student profile	Registration number, course, year of study, institution, placement facility, placement start/end dates, assigned mentor.	Avoid unnecessary personal data.
Mentor profile	Cadre, department/unit, years of practice, mentorship experience, CPD identifier if required, digital literacy self-rating.	CPD fields should be optional unless regulator-approved.
Training module	Title, objective, content type, estimated duration, quiz items, pass mark, completion status.	Modules should support text, PDF, image, video link, and quiz.
Induction checklist	Student, mentor, facility, date, orientation items, comments, completion status.	Core product asset.
Mentorship session	Date, student, mentor, session type, topic, skills discussed, feedback given, next steps, time spent.	Should take less than 3 minutes to complete.
Student feedback	Mentor availability, quality of feedback, confidence score, safety/respect score, open comments, unresolved concerns.	Provide anonymous or confidential mode depending on governance decision.
Escalation issue	Issue category, severity, description, submitted by, assigned reviewer, status, action taken, closure date.	Restrict access to authorized reviewers.
Dashboard metric	Indicator name, numerator, denominator, time period, facility, cohort, status.	Make exports easy for reports.

9. Dashboard Indicators

Indicator	Definition / calculation	Why it matters
Students registered	Number of students assigned to a facility/cohort.	Shows placement volume.
Induction completion rate	Students with completed induction checklist / total active students.	Measures structured onboarding.
Mentors onboarded	Number of mentors with active accounts.	Shows mentor reach.
Mentor training completion rate	Mentors completing all required modules / active mentors.	Measures capacity-building progress.
Mentorship sessions logged	Total and average sessions per student per week.	Shows supervision activity.
Student feedback score	Average rating across selected feedback questions.	Measures student experience.
Student confidence change	Endline confidence score minus baseline confidence score.	Measures self-reported learning growth.

Escalated issues	Open, resolved, and overdue issues by category.	Supports safeguarding and quality assurance.
Facility performance summary	Combined view of induction, sessions, training, feedback, and unresolved issues.	Supports manager and university action.

10. Suggested Training Module Structure

Module	Purpose	Suggested content
Module 1: Role of a Clinical Mentor	Clarify what mentors are expected to do.	Mentor roles, boundaries, student support, role modelling.
Module 2: Student Onboarding in Labour Ward	Standardize the first-day student experience.	Orientation checklist, unit workflow, safety, reporting lines.
Module 3: Giving Constructive Feedback	Improve quality of feedback to students.	Feedback models, respectful communication, documentation.
Module 4: Respectful Maternity Care and Learner Support	Link clinical mentorship to respectful, safe, person-centered care.	Dignity, communication, empathy, bias awareness, supportive learning environment.
Module 5: Supervision and Skills Development	Help mentors structure clinical learning.	Observation, guided practice, reflective questioning, competence discussion.
Module 6: Escalation and Safeguarding	Create a clear pathway for serious concerns.	Professional conduct, unsafe practice, harassment, confidentiality, referral steps.

11. Example Labour Ward Induction Checklist

Checklist item	Completed by	Response
Student welcomed and introduced to ward/unit team.	Mentor / nurse manager	Yes / No / Not applicable
Student shown ward layout, emergency exits, handwashing points, PPE area, and staff room.	Mentor	Yes / No / Not applicable
Student oriented to shift times, reporting lines, and who to contact for support.	Mentor	Yes / No / Not applicable
Student briefed on respectful maternity care expectations.	Mentor	Yes / No / Not applicable
Student briefed on documentation expectations and confidentiality.	Mentor	Yes / No / Not applicable
Student briefed on emergency escalation protocols.	Mentor / nurse manager	Yes / No / Not applicable
Student learning objectives discussed and recorded.	Mentor and student	Yes / No / Not applicable
First mentorship session scheduled.	Mentor and student	Date/time field

12. Technical Architecture Recommendation

The developers can choose the final stack, but the MVP should be simple, secure, and quick to deploy. A progressive web app is recommended because it supports mobile-first access without requiring installation from an app store.

Layer	Recommended approach	Developer notes
Frontend	Progressive Web App using React/Next.js, Vue, or Flutter Web.	Mobile-first forms, dashboard views, responsive UI.
Backend/API	Node.js/NestJS, Django, Laravel, or Supabase/Firebase depending on team capacity.	Use REST or GraphQL; keep business rules server-side.
Database	PostgreSQL preferred; Firebase/Supabase acceptable for rapid MVP.	Support relational reporting by facility, cohort, mentor, and student.
Authentication	Email/phone login with role-based access control.	Include password reset, invite links, and account deactivation.
Offline support	Local draft saving and retry sync for checklist/log forms.	Important for low-connectivity wards.
Dashboards	Built-in dashboard plus export to CSV/Excel; optional Looker Studio/Metabase.	Prioritize practical indicators over complex analytics.
Hosting	Secure cloud hosting or institution-approved environment.	Use HTTPS, backups, audit logs, and access controls.
Integrations	Phase 2: JKUAT virtual platform, CPD systems, WhatsApp/SMS notifications.	MVP can begin without deep integrations.

13. Privacy, Security, and Ethics Requirements

- Do not collect patient-identifiable data in the MVP.
- Include consent language for students and mentors before data collection begins.
- Use role-based access so mentors only see assigned students, while managers see only their facility or unit.
- Protect sensitive student feedback and escalation reports from unauthorized access.
- Encrypt data in transit using HTTPS and encrypt sensitive data at rest where possible.
- Include audit logs for admin actions, issue review, and data export.
- Allow account deactivation and data correction requests.
- Use anonymized or aggregated data for reporting unless identifiable reporting is explicitly approved.
- If used for research evaluation, ensure ethical approval and participant consent processes are followed.

14. Optional Phase 2 AI Features

Safety boundary

Any AI feature must support mentorship and training only. It must not provide patient-specific clinical decisions, diagnosis, prescription advice, or emergency management instructions.

AI-enabled feature	Use case	Guardrail
Mentor learning assistant	Answers questions about mentorship processes and platform use.	Use approved content only; no patient-specific advice.
Case-based scenario generator	Creates practice scenarios for mentors and students.	Scenarios must be educational and reviewed by faculty/clinical experts.

Feedback drafting helper	Helps mentors write constructive non-punitive feedback.	Mentor must review and approve all text.
Dashboard insight summaries	Summarizes trends such as low induction completion or declining feedback scores.	Use aggregate data only; avoid naming students in AI-generated summaries.
Content localization assistant	Helps simplify language or translate approved mentorship guidance.	Human review required before publishing.

15. MVP Development Backlog

Epic	User stories	Acceptance criteria
User accounts and roles	As an admin, I can invite users and assign roles. As a user, I can log in securely.	Users can log in; permissions differ by role; inactive users cannot access the system.
Facility and cohort setup	As a coordinator, I can create facilities, wards, cohorts, and placement dates.	Students and mentors can be linked to the correct facility and cohort.
Mentor training modules	As a mentor, I can complete short modules and quizzes.	System records completion, score, date, and module status.
Student onboarding	As a mentor, I can complete the labour ward induction checklist for each assigned student.	Dashboard shows each student as pending, in progress, or completed.
Mentorship session logging	As a mentor, I can quickly record each mentorship interaction.	Session log saves in under three minutes and appears in dashboard.
Student feedback	As a student, I can submit structured feedback and confidence rating.	Feedback is visible to authorized reviewers according to access rules.
Escalation workflow	As a student or mentor, I can flag serious concerns. As a manager, I can review and close issues.	Issues have severity, status, assigned reviewer, action taken, and closure date.
Dashboard and exports	As a manager/coordinator, I can view indicators and export reports.	Dashboard filters by facility, cohort, date range, and user role.

16. MVP Acceptance Criteria

- A student, mentor, nurse manager, university coordinator, and admin can each complete their core workflows without developer support.
- A mentor can complete a training module, pass a short quiz, and see completion status.
- A mentor can complete a labour ward induction checklist for a student.
- A mentor can log a mentorship session in less than three minutes.
- A student can submit feedback and confidence rating from a phone.
- A nurse manager can view incomplete inductions, inactive mentors, and unresolved issues.
- A coordinator can export data for reporting in CSV or Excel format.
- The system works on common Android phones and modern browsers.
- The system prevents ordinary users from viewing data outside their assigned role and facility.
- No patient-identifiable data is required to complete any MVP workflow.

17. Pilot Implementation Plan

Stage	Activities	Outputs
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Week 1-2: Discovery and UX	Interview mentors, students, nurse managers, and coordinators. Map existing onboarding flow and reporting needs.	User personas, workflow map, finalized fields, Figma wireframes.
Week 3-6: MVP build	Develop accounts, facility/cohort setup, training modules, induction checklist, session logs, feedback, dashboard.	Working MVP in test environment.
Week 7: Usability testing	Test with 5-10 users on mobile devices. Fix confusing screens and data capture gaps.	Revised MVP and user guide.
Week 8-16: Pilot	Run in one or two maternity units. Monitor induction, mentor engagement, feedback, and issues.	Pilot dashboard, user feedback, improvement backlog.
Endline review	Analyze user adoption, mentorship activity, student experience, and operational feasibility.	AfyaVentures pitch deck, pilot report, scale-up roadmap.

18. What Developers Need From the Project Team

- Confirmed list of pilot facilities and wards/units.
- Confirmed user roles and who should see what data.
- Final labour ward induction checklist items.
- Final mentor training module content and quiz questions.
- Decision on whether student feedback is confidential, anonymous, or identifiable to mentors.
- Consent wording for students and mentors.
- Branding assets: logo, colors, institutional names, and contact details.
- Approval on whether CPD tracking is included in MVP or later phase.
- Approval on hosting environment and data governance responsibilities.

19. Open Decisions Before Build Starts

Decision	Options	Recommended for MVP
Platform type	Web app, PWA, native Android app, native iOS app.	PWA first.
Authentication	Email, phone OTP, Google login, institutional login.	Email/phone with admin invites.
Student feedback visibility	Visible to mentor, visible to manager only, anonymous aggregate only.	Manager/coordinator view first; mentor sees aggregate or approved feedback.
CPD certification	Included now, manual certificate, regulator integration later.	Completion certificate/report now; regulator integration later.
Offline mode depth	Draft saving only, full offline sync, online only.	Draft saving and retry sync for key forms.
Languages	English only, English + Kiswahili, additional local languages.	English MVP; Kiswahili optional after user testing.
Data hosting	Institutional server, cloud, vendor-managed.	Secure cloud or JKUAT-approved environment.

20. Suggested Developer Handoff Package

The project team should provide the following package before the sprint starts:

- This concept note.
- Finalized list of data fields.
- Final labour ward induction checklist.
- Final module content for at least two MVP training modules.
- Sample student and mentor feedback forms.
- User permission matrix.
- Figma wireframes or approval to create wireframes during discovery.
- Pilot facility contacts and user testing schedule.
- Data protection and consent language.

21. Developer Summary

Build this first

A mobile-first PWA where admins create facilities/cohorts, mentors complete short digital modules, students are assigned to mentors, mentors complete induction checklists and session logs, students submit feedback, and managers/coordinators view dashboards and export reports.

MentorMAMA should be built as a practical workflow product, not just a content repository. The value is in connecting training, onboarding, mentorship activity, feedback, and dashboards into one simple system.

Appendix A: Sample Mentorship Session Log

Field	Type	Required?	Notes
Session date	Date	Yes	Default to today.
Student	Dropdown	Yes	Show only assigned students.
Mentor	Auto-filled	Yes	From logged-in user.
Session type	Dropdown	Yes	Orientation, bedside teaching, skills discussion, debrief, feedback, case reflection, other.
Topic/skill area	Multi-select	Yes	ANC, labour monitoring, documentation, newborn care, respectful maternity care, emergency preparedness, communication, other.
What was covered?	Short text	Optional	No patient names or identifiers.
Feedback given to student	Short text	Optional	Encourage constructive feedback.
Follow-up action	Short text/dropdown	Optional	None, repeat demonstration, review reading, supervisor review, other.
Time spent	Number/dropdown	Optional	Approximate minutes.

Appendix B: Sample Student Feedback Form

Question	Response type
I was properly oriented to the labour ward/unit.	1-5 scale
I know who my assigned mentor is.	Yes/No
My mentor was available when I needed guidance.	1-5 scale
I received constructive feedback on my learning.	1-5 scale
I feel more confident in my clinical learning after this week.	1-5 scale
The learning environment was respectful and supportive.	1-5 scale
Do you have any unresolved concerns that require follow-up?	Yes/No + text if yes
What is one thing that would improve your placement experience?	Open text

Appendix C: Example Permission Matrix

Feature / data	Student	Mentor	Nurse manager	University coordinator	Admin
Own profile	View/edit	View/edit	View/edit	View/edit	Manage
Facility setup	No	No	View own	Create/manage	Manage all
Student list	Own only	Assigned only	Facility only	Cohort/institution	All
Mentor training modules	View selected	Complete	View completion	View completion	Manage
Induction checklist	View own	Create/edit assigned	View facility	View cohort	Manage all
Mentorship sessions	View own summary	Create/edit own	View facility	View cohort	Manage all

Feedback	Submit own	Aggregate/approved only	Review facility	Review cohort	Manage all
Escalations	Submit/view own status	Submit/view assigned where permitted	Review facility	Review cohort	Manage all
Exports	No	No	Facility only	Cohort/institution	All

Appendix D: Short Pitch for AfyaVentures

MentorMAMA is a mobile-first mentorship platform that strengthens clinical learning in maternity units by helping mentors onboard, guide, supervise, and support nursing and midwifery students more consistently. It combines digital mentor training, labour ward induction checklists, mentorship logs, student feedback, escalation support, and dashboards for universities and facilities. The goal is to improve mentor capacity, student experience, and the quality of maternal health workforce training.